WEEK 9 · LECTURE L2 · UNIT III

Fairness, Transparency and Bias

How bias enters models, why fairness definitions conflict, and what the DPDP Act does and does not do.

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AFTER THIS LECTURE YOU SHOULD BE ABLE TO

What this lecture prepares you for

- Explain four routes by which bias enters HR models.
- Distinguish demographic parity, equalized odds and predictive parity, and explain why they conflict.
- Evaluate explainability against interpretability.
- Explain what the DPDP Act 2023 does and does not do for automated HR decisions.

COURSE OUTCOMES

CLO 3 (fairness, transparency, bias) and CLO 5.

HOW BIAS ENTERS

Four routes into a model

Route	How it works	HR example
Historical labels	The model learns from biased past decisions	Past hires that were mostly male
Proxy variables	A neutral input stands in for a protected one	Career gaps for gender; pin code for caste or class
Thin samples	Some groups are rare in the training data	Few applications from smaller towns
Feedback loops	Output shapes the next round of data	Lower allocation score, fewer orders, fewer ratings, lower score

THREE DEFINITIONS

Fair by which definition? (using the illustrative model from L1)

Definition	What must be equal across groups	Met?
Demographic parity	Share flagged	No: 50% against 20%
Equalized odds	False and true positive rates	No: FPR 47% against 18%
Predictive parity	Precision	Yes: 25% against 25%

EXAM LENS

Base rates differ (20% against 15%). When they do, no imperfect model can satisfy all three definitions at once. Choosing one is a value judgment you must defend.

What the Act does and does not do for automated decisions

Does not	Does
Regulate algorithms as such	Require accurate, complete data when data are used for decisions
Give an explicit right to an explanation	Require security safeguards and breach reporting
Let consent block staff profiling (employment is a legitimate use)	Require erasure once the purpose is served, and a grievance route

EXAM LENS

Under the DPDP Act, the audit, human review and a working grievance route are the real safeguards, not consent.

THEORY IN DEPTH · KLEINBERG ET AL. (2016); CHOULDECHOVA (2017)

Fairness definitions cannot all hold at once

THE CORE IDEA

When groups have different base rates, a model cannot be calibrated and have equal error rates across groups, unless it is perfect.

HOW IT WORKS

Improving one fairness measure necessarily worsens another.

LIMITS AND CRITIQUE

The result is mathematical; deciding which fairness matters is an ethical and legal choice.

IN PRACTICE

An employer using a flight-risk model must state which definition it chose, and why.

EXAM LENS

This impossibility result is a powerful point in any critical answer on algorithmic fairness.

THEORY IN DEPTH · BUOLAMWINI AND GEBRU (2018)

Bias hides at the intersections

THE CORE IDEA

The Gender Shades study tested commercial face-analysis systems on gender classification.

HOW IT WORKS

Error rates reached 34.7% for darker-skinned women against 0.8% for lighter-skinned men.

LIMITS AND CRITIQUE

One study of one task, but it changed industry practice and showed why single-axis audits miss harm.

IN PRACTICE

A model fair by gender and fair by age can still be unfair to young women. Audit the intersections.

EXAM LENS

Cite the exact figures; they make the intersectionality argument concrete.

THEORY IN DEPTH · RUDIN (2019)

Use interpretable models for high-stakes decisions

THE CORE IDEA

Rather than explaining black boxes after the fact, build models people can understand directly.

HOW IT WORKS

In many tabular problems, simple scoring models match black-box accuracy.

LIMITS AND CRITIQUE

Some tasks, such as images and free text, still need complex models.

IN PRACTICE

For most HR decisions on tabular data, a transparent scoring rule is likely to be as accurate as a black box.

EXAM LENS

Distinguish explainability (post hoc) from interpretability (by design); Rudin argues for the latter.

THEORY IN DEPTH · LEICHT-DEOBALD ET AL. (2019)

Algorithms can crowd out personal integrity

THE CORE IDEA

When systems decide, people may stop exercising moral judgment and simply comply.

HOW IT WORKS

Algorithm-based HR decisions can marginalize human sense-making, encourage blind trust in rules, and dull moral imagination.

LIMITS AND CRITIQUE

Proposed remedies include critical data literacy, ethical awareness and keeping humans accountable for decisions.

IN PRACTICE

A manager who says "the system flagged her" has handed away a judgment that remains his.

EXAM LENS

Use this to argue that human review must carry real accountability, not just sign-off.

PRACTICE FOR THE END-TERM EXAMINATION

Questions to rehearse

"There is no such thing as a fair algorithm, only a chosen definition of fairness." Critically discuss using the impossibility results and an HR example. **20 marks**

A strong answer: defines the three criteria; explains the base-rate conflict with numbers; argues the choice is normative; links to accountability.

Evaluate the adequacy of the DPDP Act 2023 for governing algorithmic HR decisions. **20 marks**

A strong answer: states what the Act does and does not do; compares with consent-based and explanation-based approaches; proposes organizational safeguards.

Distinguish explainability from interpretability, with reference to Rudin (2019). **10 marks**

A strong answer: gives precise definitions and Rudin's argument.

CARRY THESE INTO THE EXAMINATION

Three things to remember

1. **Bias enters through labels, proxies, thin samples and feedback loops**
2. **Fairness definitions conflict; choose one and defend it**
3. **Under the DPDP Act, audit and redress, not consent, protect staff**

SET READINGS · FROM THE LESSON PLAN

Read for this lecture

- Leicht-Deobald, U., et al. (2019). *Journal of Business Ethics, 160*(2), 377–392.
- The DPDP employee data note.

COMPANION

Each set reading is explained in the matching Unit Reader on rksingh.tech/materials.

THEORY SOURCES · FOR REFERENCE

Sources for this lecture

Works named on the slides of this lecture. They are background sources, not set readings.

Buolamwini, J., & Gebru, T. (2018). Gender shades: Intersectional accuracy disparities in commercial gender classification. *Proceedings of Machine Learning Research, 81*, 77–91.

Chouldechova, A. (2017). Fair prediction with disparate impact: A study of bias in recidivism prediction instruments. *Big Data, 5*(2), 153–163. https://doi.org/10.1089/big.2016.0047

Kleinberg, J., Mullainathan, S., & Raghavan, M. (2016). *Inherent trade-offs in the fair determination of risk scores* (arXiv:1609.05807). arXiv. https://doi.org/10.48550/arXiv.1609.05807

Leicht-Deobald, U., Busch, T., Schank, C., Weibel, A., Schafheitle, S., Wildhaber, I., & Kasper, G. (2019). The challenges of algorithm-based HR decision-making for personal integrity. *Journal of Business Ethics, 160*(2), 377–392. https://doi.org/10.1007/s10551-019-04204-w

Rudin, C. (2019). Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead. *Nature Machine Intelligence, 1*(5), 206–215. https://doi.org/10.1038/s42256-019-0048-x